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INT305 Machine Learning

Lecture 3

Linear Classifiers, Logistic Regression, Multiclass Classification

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Overview

- **Classification**: predicting a discrete-valued target
 - **Binary classification**: predicting a binary-valued target
 - **Multiclass classification**: predicting a discrete(>2)-valued target
- Examples of binary classification
 - Predict whether a patient has a disease, given the presence or absence of various symptoms
 - Classify e-mails as spam or non-spam
 - Predict whether a financial transaction is fraudulent

Overview

Binary linear classification

- **Classification:** given a D -dimensional input $\mathbf{x} \in \mathbb{R}^D$ predict a discrete-valued target
- **Binary:** predict a binary target $t \in \{0,1\}$
 - Training examples with $t = 1$ are called **positive examples**, and training examples with $t = 0$ are called **negative examples**. Sorry.
 - $t \in \{0,1\}$ or $t \in \{-1, +1\}$ is for computational convenience.
- **Linear:** model prediction y is a linear function of $\mathbf{x}$, followed by a threshold r :

$$z = \mathbf{w}^\top \mathbf{x} + b$$

$$y = \begin{cases} 1 & \text{if } z \geq r \\ 0 & \text{if } z < r \end{cases}$$

Simplifications

Eliminating the threshold

- We can assume without loss of generality (WLOG) that the threshold $r = 0$:

$$\mathbf{w}^\top \mathbf{x} + b \geq r \iff \mathbf{w}^\top \mathbf{x} + \underbrace{b - r}_{\triangleq w_0} \geq 0$$

Eliminating the bias

- Add a dummy feature x_0 which always takes the value 1. the weight $w_0 = b$ is equivalent to a bias (same as linear regression)

Simplified model

- Receive input $\mathbf{x} \in \mathbb{R}^{D+1}$ with $x_0 = 1$:

$$z = \mathbf{w}^\top \mathbf{x}$$

$$y = \begin{cases} 1 & \text{if } z \geq 0 \\ 0 & \text{if } z < 0 \end{cases}$$

简化

Examples

- Let's consider some simple examples to examine the properties of our model
- Let's focus on minimizing the training set error, and forget about whether our model will generalize to a test set.

Examples

NOT

bias 项

x_0	x_1	t
1	0	1
1	1	0

- Suppose this is our training set, with the dummy feature x_0 included.
- Which conditions on w_0, w_1 guarantee perfect classification?
 - When $x_1 = 0$, need: $z = w_0x_0 + w_1x_1 \geq 0 \iff w_0 \geq 0$
 - When $x_1 = 1$, need: $z = w_0x_0 + w_1x_1 < 0 \iff w_0 + w_1 < 0$
- Example solution: $w_0 = 1, w_1 = -2$
- Is this the only solution?

✗

找到一组解

Examples

AND

x_0	x_1	x_2	t
1	0	0	0
1	0	1	0
1	1	0	0
1	1	1	1

$$z = w_0x_0 + w_1x_1 + w_2x_2$$

$$\text{Need: } w_0 < 0$$

$$\text{Need: } w_0 + w_2 < 0$$

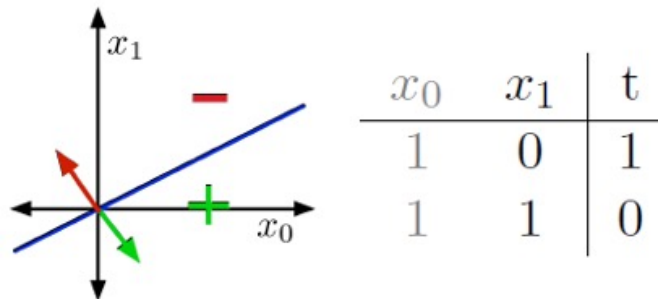
$$\text{Need: } w_0 + w_1 < 0$$

$$\text{Need: } w_0 + w_1 + w_2 \geq 0$$

Example solution: $w_0 = -1.5, w_1 = 1, w_2 = 1$

The Geometric Picture

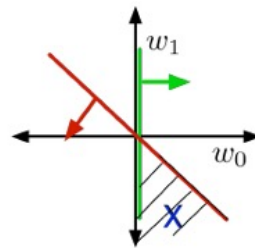
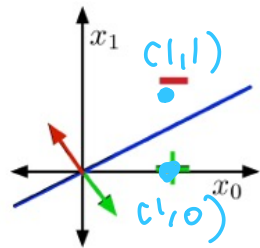
Input Space, or Data Space for NOT example



- Training examples are points
- Weights (hypotheses) $\mathbf{w}$ can be represented by half-spaces
 $H_+ = \{\mathbf{x}: \mathbf{w}^\top \mathbf{x} \geq 0\}$, $H_- = \{\mathbf{x}: \mathbf{w}^\top \mathbf{x} < 0\}$
 - The boundaries of these half-spaces pass through the origin (why?)
- The boundary is the decision boundary: $\{\mathbf{x}: \mathbf{w}^\top \mathbf{x} = 0\}$
 - In 2-D, it's a line, but in high dimensions it is a hyperplane
- If the training examples can be perfectly separated by a linear decision rule, we say data is linearly separable.

The Geometric Picture

Weight Space



$$w_0 \geq 0$$

$$w_0 + w_1 < 0$$

- Weights (hypotheses) $\mathbf{w}$ are points
- Each training example $\mathbf{x}$ specifies a half-space $\mathbf{w}$ must lie in to be correctly classified:
 $\mathbf{w}^T \mathbf{x} \geq 0$ if $t = 1$.
- For NOT example:
 - $x_0 = 1, x_1 = 0, t = 1 \implies (w_0, w_1) \in \{\mathbf{w}: w_0 \geq 0\}$
 - $x_0 = 1, x_1 = 1, t = 0 \implies (w_0, w_1) \in \{\mathbf{w}: w_0 + w_1 < 0\}$
- The region satisfying all the constraints is the **feasible region**; if this region is nonempty, the problem is **feasible**, otw it is **infeasible**.

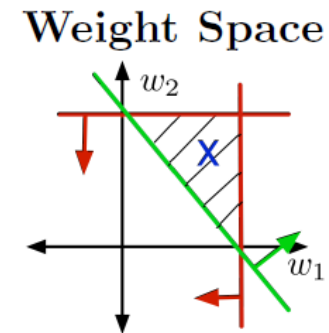
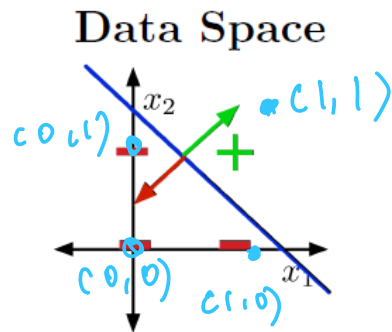
可行的

The Geometric Picture

- The **AND** example requires three dimensions, including the dummy one.
- To visualize data space and weight space for a 3-D example, we can look at a 2-D slice.
- The visualizations are similar.
 - Feasible set will always have a corner at the origin.

The Geometric Picture

Visualizations of the **AND** example



- Slice for $x_0 = 1$ and
- Example sol: $w_0 = -1.5, w_1 = 1, w_2 = 1$
- Decision boundary:

$$w_0x_0 + w_1x_1 + w_2x_2 = 0$$

$$\implies -1.5 + x_1 + x_2 = 0$$

- Slice for $w_0 = -1.5$ for the constraints
- $w_0 < 0$
- $w_0 + w_2 < 0$
- $w_0 + w_1 < 0$
- $w_0 + w_1 + w_2 \geq 0$

Summary | Binary Linear Classifiers

- **Summary:** targets $t \in \{0,1\}$, inputs $\mathbf{x} \in \mathbb{R}^{D+1}$ with $x_0 = 1$, and model is defined by weights $\mathbf{w}$ and

$$z = \mathbf{w}^\top \mathbf{x}$$

$$y = \begin{cases} 1 & \text{if } z \geq 0 \\ 0 & \text{if } z < 0 \end{cases}$$

- How can we find good values for $\mathbf{w}$?
- If training set is linearly separable, we could solve for $\mathbf{w}$ using linear programming
 - We could also apply an iterative procedure known as the *perceptron algorithm* (but this is primarily of historical interest).
- If it's not linearly separable, the problem is harder
 - Data is almost never linearly separable in real life.

Towards Logistic Regression

Towards Logistic Regression 逻辑回归

Loss Functions

- Instead: define loss function then try to minimize the resulting cost function
 - Recall : cost is loss averaged (or summed) over the training set
- Seemingly obvious loss function: **0-1 loss**

$$\begin{aligned}\mathcal{L}_{0-1}(y, t) &= \begin{cases} 0 & \text{if } y = t \\ 1 & \text{if } y \neq t \end{cases} \\ &= \mathbb{I}[y \neq t]\end{aligned}$$

Attempt 1: 0-1 loss

- Usually, the cost $\mathcal{J}$ is the averaged loss over training examples; for 0-1 loss, this is the **misclassification rate**:

$$\mathcal{J} = \frac{1}{N} \sum_{i=1}^N \mathbb{I}[y^{(i)} \neq t^{(i)}]$$

Attempt 1: 0-1 loss

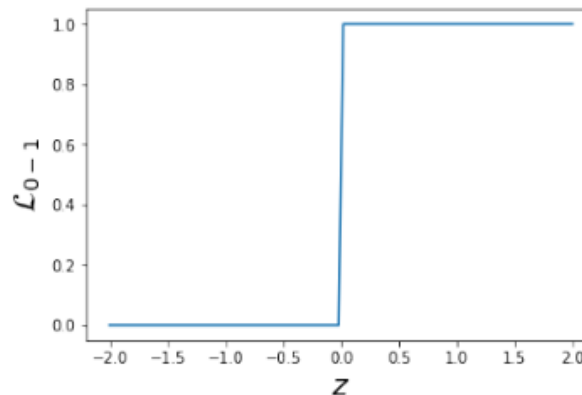
- Problem: how to optimize? In general, a hard problem (can be NP-hard)
- This is due to the step function (0-1 loss) not being nice (continuous/smooth/convex etc)

Attempt 1: 0-1 loss

- Minimum of a function will be at its critical points.
- Let's try to find the critical point of 0-1 loss.
- Chain rule:

$$\frac{\partial \mathcal{L}_{0-1}}{\partial w_j} = \frac{\partial \mathcal{L}_{0-1}}{\partial z} \frac{\partial z}{\partial w_j}$$

- But $\partial \mathcal{L}_{0-1} / \partial z$ is zero everywhere it's defined!



- $\partial \mathcal{L}_{0-1} / \partial w_j = 0$ means that changing the weights by a very small amount probably has no effect on the loss.
- Almost any point has 0 gradient!

Attempt 2: Linear Regression

- Sometimes we can replace the loss function we care about with one which is easier to optimize. This is known as **relaxation** with a smooth **surrogate loss function**.
- One problem with $\mathcal{L}_{0-1}$: defined in terms of final prediction, which inherently involves a discontinuity.
- Instead, define loss in terms of $\mathbf{w}^\top \mathbf{x}$ directly.
 - Redo notation for convenience: $z = \mathbf{w}^\top \mathbf{x}$

Attempt 2: Linear Regression

- We already know how to fit a linear regression model. Can we use this instead?

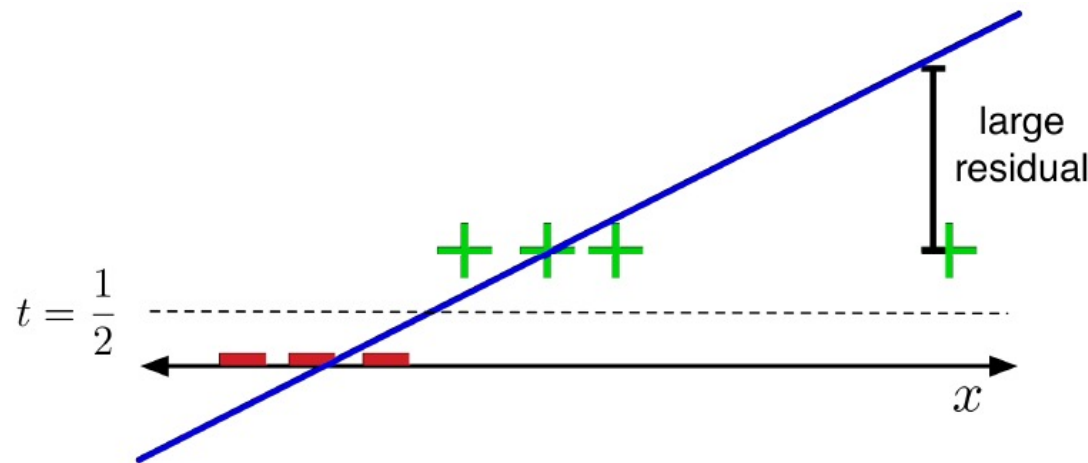
$$z = \mathbf{w}^\top \mathbf{x}$$

$$\mathcal{L}_{\text{SE}}(z, t) = \frac{1}{2} (z - t)^2$$

- Doesn't matter that the targets are actually binary. Treat them as continuous values.
- For this loss function, it makes sense to make final predictions by thresholding z at $\frac{1}{2}$ (why?)

Attempt 2: Linear Regression

The problem:



- The loss function hates when you make correct predictions with high confidence!

- It $t = 1$, it's more unhappy about $z = 10$ than $z = 0$.

这里是说用线性回归用在分类问题上， $z=10$ 和 0 的标签明明只是标签，却被用于预测值和预测值差异

Attempt 3: Logistic Activation Function

- There's obviously no reason to predict values outside $[0, 1]$. Let's squash y into this interval.
- The **logistic function** is a kind of **sigmoid**, or S-shaped function:

$$\sigma(z) = \frac{1}{1 + e^{-z}}$$

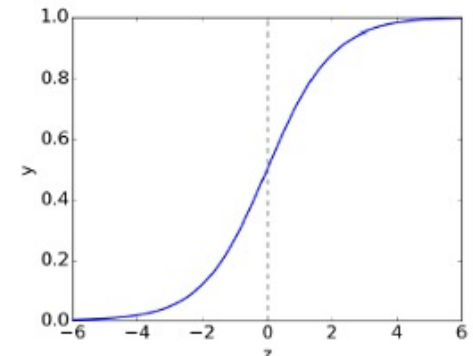
- $\sigma^{-1}(y) = \log(y/(1 - y))$ is called the **logit**.
- A linear model with a logistic nonlinearity is known as **log-linear**:

$$z = \mathbf{w}^\top \mathbf{x}$$

$$y = \sigma(z)$$

$$\mathcal{L}_{\text{SE}}(y, t) = \frac{1}{2} (y - t)^2$$

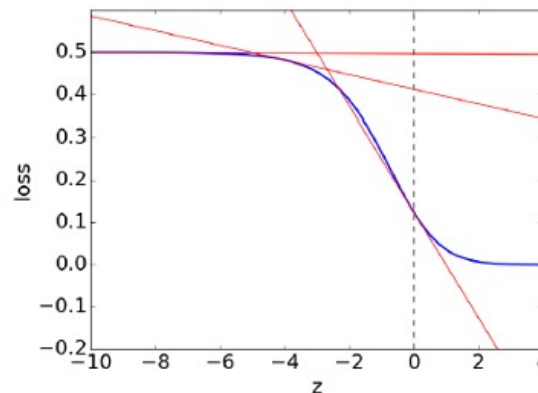
- Used in this way, σ is called an **activation function**.



Attempt 3: Logistic Activation Function 的问题(如果损失函数是SE)

The problem:

(plot of $\mathcal{L}_{SE}$ as a function of z , assuming $t = 1$)



$$\frac{\partial \mathcal{L}}{\partial w_j} = \frac{\partial \mathcal{L}}{\partial z} \frac{\partial z}{\partial w_j}$$

- For $z \ll 0$, we have $\sigma(z) \approx 0$.
- $\frac{\partial \mathcal{L}}{\partial z} \approx 0$ (check) $\rightarrow \frac{\partial \mathcal{L}}{\partial w_j} \approx 0 \rightarrow$ derivative w.r.t. w_j is small $\rightarrow w_j$ is like a critical point.
- If the prediction is really wrong, you should be far from a critical point (which is your candidate solution).

真实标签是1, 模型预测概率为0; 应该让参数 update, 由于 sigmoid

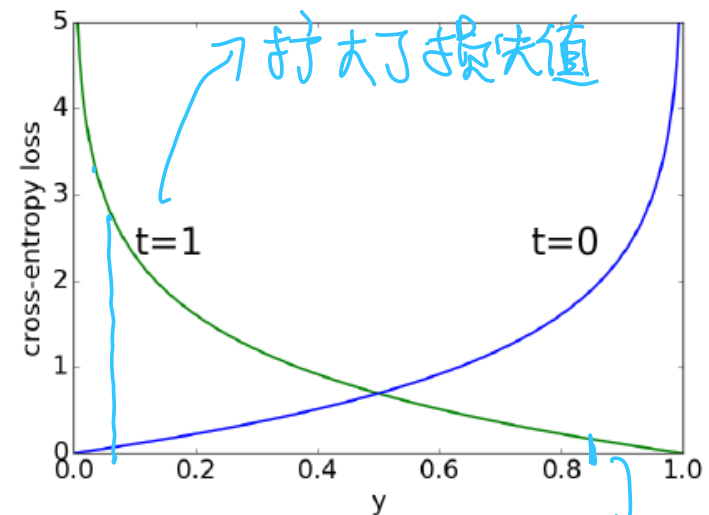
在两侧导数都为0, 无法修正问题

Logistic Regression

- Because $y \in [0, 1]$, we can interpret it as the estimated probability that $t = 1$. If $t = 0$, then we want to heavily penalize $y \approx 1$.
- ~~X~~ The pundits who were 99% confident Clinton would win were much more wrong than the ones who were only 90% confident. (别看例子)
- **Cross-entropy loss** (aka log loss) captures this intuition:

$$\mathcal{L}_{\text{CE}}(y, t) = \begin{cases} -\log y & \text{if } t = 1 \\ -\log(1 - y) & \text{if } t = 0 \end{cases}$$

$$= -t \log y - (1 - t) \log(1 - y)$$



预测准

对于MSE问题修正

使用 Cross-Entropy Loss (分情况计算损失)

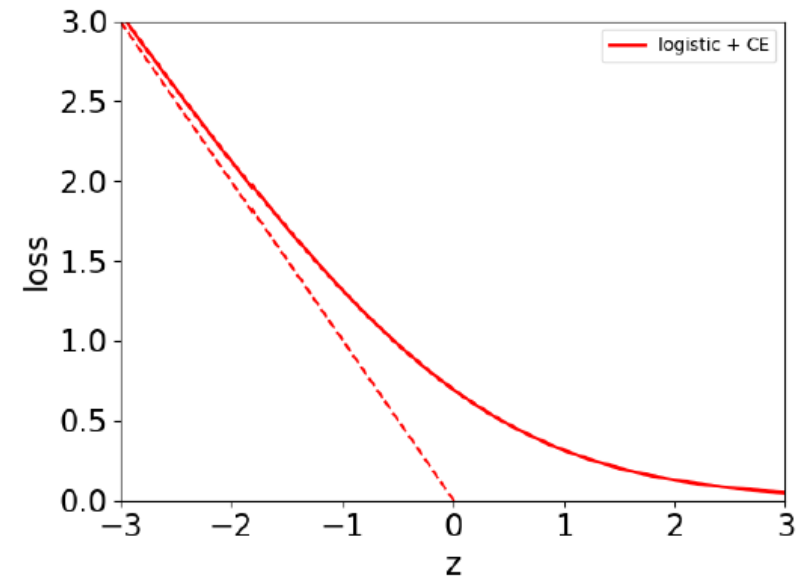
Logistic Regression

Logistic regression:

$$z = \mathbf{w}^T \mathbf{x}$$
$$y = \sigma(z)$$
$$= \frac{1}{1 + e^{-z}}$$

$$\mathcal{L}_{\text{CE}}(y, t) = -t \log y - (1 - t) \log(1 - y)$$

Plot is for target $t = 1$.



Gradient Descent for Logistic Regression

- How do we minimize the cost $\mathcal{J}$ for logistic regression? No direct solution.
 - Taking derivatives of $\mathcal{J}$ w.r.t. $\mathbf{w}$ and setting them to 0 doesn't have an explicit solution.
- However, the logistic loss is a **convex function** in $\mathbf{w}$, so let's consider the **gradient descent** method from the last lecture.
 - Recall: we **initialize** the weights to something reasonable and repeatedly adjust them in the **direction of steepest descent**.
 - A standard initialization is $\mathbf{w} = 0$. (why?)

Gradient of Logistic Loss

$y \rightarrow \text{pred}$ $t \rightarrow \text{true data}$

Back to logistic regression:

$$\mathcal{L}_{\text{CE}}(y, t) = -t \log y - (1 - t) \log(1 - y)$$

$$y = 1 / (1 + e^{-z}) \text{ and } z = \mathbf{w}^T \mathbf{x}$$

Therefore

$$\begin{aligned} \frac{\partial \mathcal{L}_{\text{CE}}}{\partial w_j} &= \frac{\partial \mathcal{L}_{\text{CE}}}{\partial y} \cdot \frac{\partial y}{\partial z} \cdot \frac{\partial z}{\partial w_j} = \left(-\frac{t}{y} + \frac{1-t}{1-y} \right) \cdot y(1-y) \cdot x_j \\ &= (y - t)x_j \end{aligned}$$

(verify this)

Gradient descent (coordinatewise) update to find the weights of logistic regression:

$$\begin{aligned} w_j &\leftarrow w_j - \alpha \frac{\partial \mathcal{J}}{\partial w_j} \\ &= w_j - \frac{\alpha}{N} \sum_{i=1}^N (y^{(i)} - t^{(i)}) x_j^{(i)} \end{aligned}$$

Gradient Descent for Logistic Regression

Comparison of gradient descent updates:

- Linear regression:

$$\mathbf{w} \leftarrow \mathbf{w} - \frac{\alpha}{N} \sum_{i=1}^N (y^{(i)} - t^{(i)}) \mathbf{x}^{(i)}$$

- Logistic regression:

$$\mathbf{w} \leftarrow \mathbf{w} - \frac{\alpha}{N} \sum_{i=1}^N (y^{(i)} - t^{(i)}) \mathbf{x}^{(i)}$$

为啥一致: 由于都是广义线性模型

- Not a coincidence! These are both examples of **generalized linear models**. But we won't go in further detail.
- Notice $\frac{1}{N}$ in front of sums due to averaged losses. This is why you need smaller learning rate when cost is summed losses ($\alpha' = \alpha/N$).

Multiclass Classification

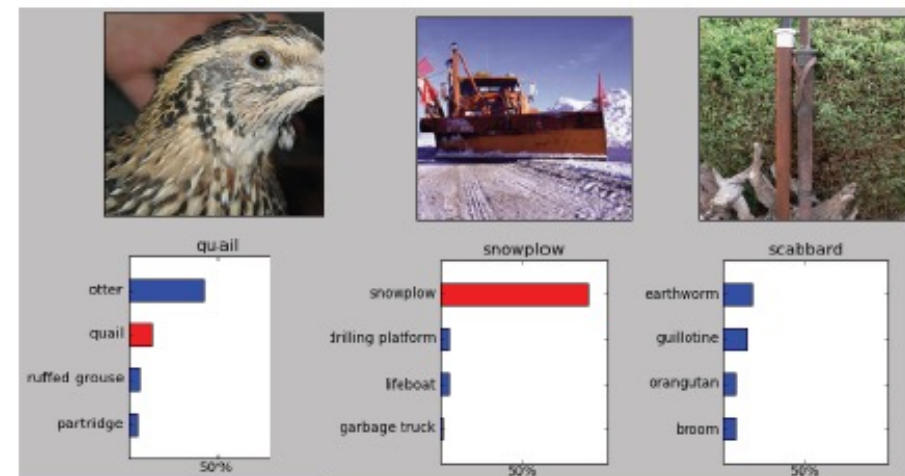
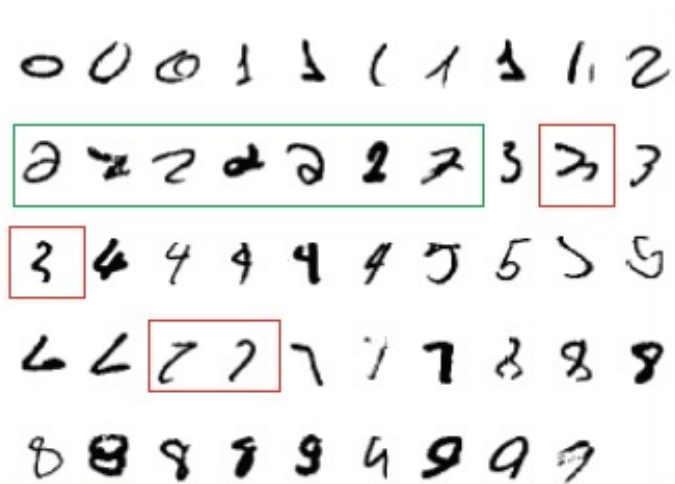
Multiclass Classification and Softmax Regression

Overview

- **Classification**: predicting a discrete-valued target
 - **Binary classification**: predicting a binary-valued target
 - **Multiclass classification**: predicting a discrete-(>2)-valued target
- Examples of multi-class classification
 - Predict the value of a handwritten digit
 - Classify e-mails as spam, travel, work, personal

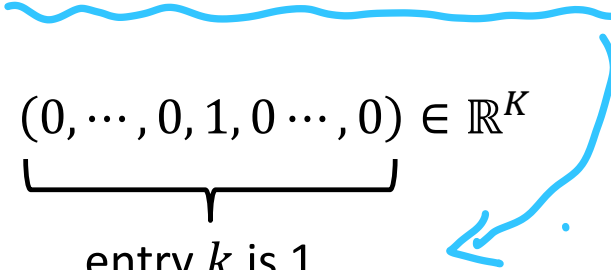
Multiclass Classification

- Classification tasks with more than two categories:



Multiclass Classification

- Targets form a discrete set $\{1, \dots, K\}$.
- It's often more convenient to represent them as one-hot vectors, or a one-of-K encoding:

$$\mathbf{t} = (\underbrace{0, \dots, 0, 1, 0, \dots, 0}_{\text{entry } k \text{ is } 1}) \in \mathbb{R}^K$$


Multiclass Linear Classification

- We can start with a linear function of the inputs.
- Now there are D input dimensions and K output dimensions, so we need $K \times D$ weights, which we arrange as a **weight matrix $\mathbf{W}$** .
- Also, we have a K -dimensional vector $\mathbf{b}$ of biases.
- A linear function of the inputs:

$$z_k = \sum_{j=1}^D w_{kj} x_j + b_k \quad \text{for } k = 1, 2, \dots, K$$

- We can eliminate the bias $\mathbf{b}$ by taking $\mathbf{W} \in \mathbb{R}^{K \times (D+1)}$ and adding a dummy variable $x_0 = 1$. So, vectorized:

$$\mathbf{z} = \mathbf{W}\mathbf{x} + \mathbf{b} \quad \text{or with dummy } x_0 = 1 \quad \mathbf{z} = \mathbf{W}\mathbf{x}$$

Multiclass Linear Classification

- How can we turn this linear prediction into a **one-hot prediction**?
- We can interpret the magnitude of z_k as a measure of how much the model prefers k as its prediction.
- If we do this, we should set

$$y_i = \begin{cases} 1 & i = \arg \max_k z_k \\ 0 & \text{otherwise} \end{cases}$$

$$y = \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_N \end{pmatrix} \leftarrow y_k = \frac{e^{z_k}}{\sum_k e^{z_k}} \leftarrow z = Wx$$

Softmax Regression

- We need to soften our predictions for the sake of optimization.
- We want soft predictions that are like probabilities, i.e., $0 \leq y_k \leq 1$ and $\sum_k y_k = 1$.
- A natural activation function to use is the **softmax function**, a multivariable generalization of the logistic function:

$$y_k = \text{softmax}(z_1, \dots, z_K)_k = \frac{e^{z_k}}{\sum_{k'} e^{z_{k'}}}$$

- Outputs can be interpreted as probabilities (positive and sum to 1)
- If z_k is much larger than the others, then $\text{softmax}(\mathbf{z})_k \approx 1$ and it behaves like argmax.

Softmax Regression

- If a model outputs a vector of class probabilities, we can use cross-entropy as the loss function:

$$\begin{aligned}\mathcal{L}_{\text{CE}}(\mathbf{y}, \mathbf{t}) &= - \sum_{k=1}^K t_k \log y_k \\ &= -\mathbf{t}^T (\log \mathbf{y}),\end{aligned}$$

where the log is applied elementwise.

- Just like with logistic regression, we typically combine the softmax and cross-entropy into a **softmax-cross-entropy** function.

Softmax Regression

- Softmax regression (with dummy $x_0 = 1$):

$$\mathbf{z} = \mathbf{W}\mathbf{x}$$

$$\mathbf{y} = \text{softmax}(\mathbf{z})$$

$$\mathcal{L}_{\text{CE}} = -\mathbf{t}^\top (\log \mathbf{y})$$

- Gradient descent updates can be derived for each row of $\mathbf{W}$:

$$\frac{\partial \mathcal{L}_{\text{CE}}}{\partial \mathbf{w}_k} = \frac{\partial \mathcal{L}_{\text{CE}}}{\partial z_k} \cdot \frac{\partial z_k}{\partial \mathbf{w}_k} = (y_k - t_k) \cdot \mathbf{x}$$

$$\mathbf{w}_k \leftarrow \mathbf{w}_k - \alpha \frac{1}{N} \sum_{i=1}^N \left(y_k^{(i)} - t_k^{(i)} \right) \mathbf{x}^{(i)}$$

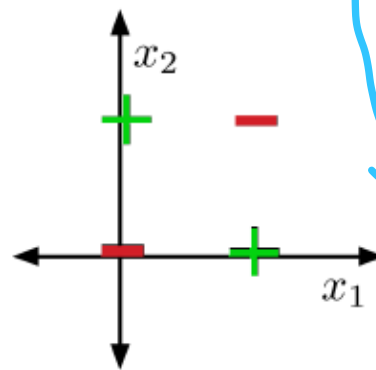
- Similar to linear/logistic reg (no coincidence) (verify the update)

Prove the gradient ?

Limits of Linear Classification

Some datasets are not linearly separable, e.g. **XOR**

有些数据集不具备线性可分性



XOR

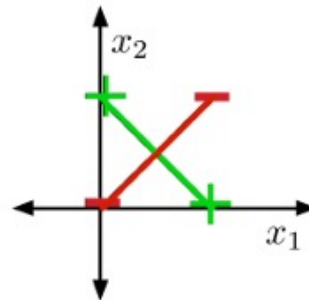
0, 0	→ 0
0, 1	→ 1
1, 0	→ 1
1, 1	→ 0

Visually obvious, but how to show this?

Limits of Linear Classification

Showing that XOR is not linearly separable (proof by contradiction)

- If two points lie in a half-space, line segment connecting them also lie in the same half-space.
- Suppose there were some feasible weights (hypothesis). If the positive examples are in the positive half-space, then the green line segment must be as well.
- Similarly, the red line segment must lie within the negative half-space.



- But the intersection can't lie in both half-spaces. Contradiction!

Limits of Linear Classification

- Sometimes we can overcome this limitation using **feature maps**, just like for linear regression. E.g., for **XOR**:

无解
 $w_2 > 0$
 $w_1 > 0$
 $w_1 + w_2 \leq 0$

$$\psi(\mathbf{x}) = \begin{pmatrix} x_1 \\ x_2 \\ x_1 x_2 \end{pmatrix}$$

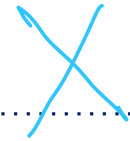
x_1	x_2	$\psi_1(\mathbf{x})$	$\psi_2(\mathbf{x})$	$\psi_3(\mathbf{x})$	t
0	0	0	0	0	0
0	1	0	1	0	1
1	0	1	0	0	1
1	1	1	1	1	0

$w_2 > 0$
 $w_1 > 0$
 $w_1 + w_2 + w_3 \leq 0$
 有解

- This is linearly separable. (Try it!)

通过特征映射使得线性可分

Next time...



Feature maps are hard to design well, so next time we'll see how to *learn* nonlinear feature maps directly using neural networks...

